

20/08/2025

Task-3

Importing and creating python modulus and packages in python program

Aim: To implement and demonstrate the process of importing built-in modulus, creating user defined modulus, and organizing code into packages in python, thereby promoting code reusability, modularity and maintainability.

3.1

- 1) perform common math and random operations.
- 2) Work with the Operating System (create/change directories, list contents) and read the python version.
- 3) Compute basic statistics (mean, median, mode, standard deviation).

Algorithm:

- 1) Input required modulus: math, random, os, sys, statistics, pathlib.
- 2) Math & random:
 - compute $\sqrt{5}$, radians(30), a random float in $[0.0, 1.0)$, a random integer in $[2, 6]$ (inclusive), π , ceil(2.3), floor(2.5), factorial(5), gcd(5, 15), abs(-10), pow(3, 5), log base 3 of 2, $\log_{10}(a)$ for $a=100$, and check the nan/infinity.
- 3) OS and sys:
 - print python interpreter version
- 4) Statistics:
 - on lists: $[5, 6, 8, 10]$ and $[2, 5, 3, 2, 8, 3, 9, 4, 2, 5, 6]$, compute mean, median, mode, stdev.
- 5) print neatly formatted results.

Program:

```
import math
import random
import os
import sys
import statistics as stats
from pathlib import Path
print("\n --- MATH & RANDOM ---")
```


output:

----- MATH & RANDOM -----

$$\text{sqrt}(5) = 2.23606797749979$$

$$\text{radians}(30) = 0.523598$$

$$\text{random}() \text{ in } [0, 1) = 0.374448871$$

$$\text{randint}(2, 6) = 6$$

$$\pi = 3.141592$$

$$\text{ceil}(2.3) = 3$$

$$\text{floor}(2.3) = 2$$

$$\text{factorial}(5) = 120$$

$$\text{gcd}(5, 15) = 5$$

$$\text{abs}(-10) = 10$$

$$\text{pow}(3, 5) = 243$$

----- OS & SYS -----

Created/ensured : c:\python\lab

current working directory : c:\python\lab (your current path)

Created/ensured & changed

Directory contents of c:\python\lab

python version: 3.x.x

----- STATISTICS -----

$$\text{mean}([5, 6, 8, 10]) = 7.25$$

$$\text{median}([5, 6, 8, 10]) = 7.0$$

$$\text{mode}([2, 5, 3, 2, 8, 3, 9, 4, 2, 5, 6]) = 2$$

$$\text{stddev}([2, 5, 3, 2, 8, 3, 9, 4, 2, 5, 6]) = 2.271563338$$


```

print("sqrt(5) =", math.sqrt(5))
print("radians(30) =", math.radians(30))
print("random() in [0,1) =", random.random())
print("randint(2,6) =", random.randint(2,6)) # inclusive
print("pi =", math.pi)
print("ceil(2.3) =", math.ceil(2.3))
print("floor(2.3) =", math.floor(2.3))
print("factorial(5) =", math.factorial(5))
print("gcd(5,15) =", math.gcd(5,15))
print("abs(-10) =", abs(-10))
print("pow(3,5) =", pow(3,5))
print("log base 3 of 2 =", math.log(2,3))
a_val = 100
print(f"isinf(∞) = {math.isinf(inf_val)} isnan(NaN) = {math.isnan(a_val)}")
print("\n--- os < sys ---")
path-pythonlab = path('r"c:\pythonlab')
path-pythonlab.mkdir(parents=True, exist_ok=True)
print(f"created/ensured '{path-pythonlab}'")
print("Current working directory =", os.getcwd())
target_dir = path('r"c:\pythonlab\5214')
target_dir.mkdir(parents=True, exist_ok=True)
os.chdir(target_dir)
print("\n--- STATISTICS ---")
data_1 = [5, 6, 8, 10]
data_2 = [2, 5, 3, 2, 8, 3, 9, 4, 2, 5, 6]
print(f"mean({data_1}) =", stats.mean(data_1))
print(f"median({data_1}) =", stats.median(data_1))
print(f"mode({data_2}) =", stats.mode(data_2))
print(f"stdev({data_2}) =", stats.stdev(data_2))

```


3.2 Create a python package named cardpack containing a module cardfun that imports the random module. Assign a range of cards, call a function from the module, and display a random sample of cards.

Algorithm:

Step 1: Start

Step 2: To create a package cardpack

Step 3: To create a module cardfun and import random function

Step 4: Assign a cards range

Step 5: Call a module function.

Step 6: Display the random sample card Step 7: stop

Program:

Card Fun

```
import random
```

```
def func():
```

```
    cards: []
```

```
    for i in range(1, 53):
```

```
        cards.append(i)
```

```
    shuffled_cards = random.sample(cards, k=52)
```

```
    print("\n\n", shuffled_cards, "\n\n")
```

my mod.py

```
import cardFun
```

```
cardFun.func()
```


Output:

RESTART:

C:\users\student.MAT2RC6833\AppData\local

{5, 24, 13, 22, 20, 41, 48, 38, 51, 14, 7, 34, 49, 14, 50,
37, 40, 15, 35, 17, 18, 33, 39, 36, 42, 12, 6, 16, 19, 48,
29, 2, 27, 11, 31, 36, 28, 21, 32, 8, 25, 30, 23, 12, 6, 10,
43, 44, 31, 44, 52, 1, 49, 9}

3.3 : You are tasked with developing a modular calculator application in python. The calculator should support basic arithmetic operations should be implemented in a separate module.

Algorithm:

- 1) Define functions for addition, subtraction, multiplication and division
- 2) Handle division by zero by raising an error if the divisor is zero.
- 3) Import the module
- 4) Initialize two ~~new~~ numbers ($a=10, b=5$).
- 5) Call each function using mymath.
- 6) print the results of all operations.

Program:

```
def add(a,b):  
    return a+b  
def subtract(a,b):  
    return a-b  
def multiply(a,b):  
    return a*b  
def divide(a,b):  
    if b==0:  
        raise ValueError("Cannot divide by zero")  
    return a/b  
  
import mymath.  
a=10  
b=5  
print("Addition :", mymath.add(a,b))  
print("Subtraction :", mymath.subtract(a,b))  
print("Multiplication :", mymath.multiply(a,b))  
print("Division :", mymath.divide(a,b))
```


3.4 You are working on a python project that requires you to perform various mathematical operations and geometric area calculations. To organize your code better, you decide to create a package named my package which includes sub packages pack1 and pack2.

Algorithm:

- 1) Create math functions.py module
- 2) Create area functions.py module:
- 3) Create main.py:
- 4) print the output as expected.

Program:

1. Create the math functions.py module.

```
def add(a,b):  
    return a+b  
def subtract(a,b):  
    return a-b  
def multiply(a,b):  
    return a*b  
def divide(a,b):  
    if b==0:  
        return "Error: Division by Zero".  
    return a/b
```

2. create the area functions.py module.

```
import math  
def circle_area(radius):  
def rectangle_area(length, width):  
    return length * width  
def triangle_area(base, height):  
    return 0.5 * base * height
```

3. Create the main.py file

```
import mathfunctions  
import areafunctions  
# using math functions  
print("Addition :", mathfunctions.add(10, 5))
```



```
print("subtraction:", mathfunctions.subtract(10,5))
```

```
print("Multiplication :", mathfunctions.multiply(10,5))
```

```
print("Division :", mathfunctions.divide(10,5))
```

#using area functions.

```
print("Circle Area(radius=7):", areafunctions.circle_area(7))
```

```
print("Rectangle Area(5x10):", areafunctions.rectangle_area(5,10))
```

```
print("Triangle Area(base=6, height=8):", areafunctions.triangle_area(6,8))
```

Result: This program for importing Python modules and package was successfully executed and the output was verified.

VEL TECH	
EX NO.	2
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	
TOTAL (20)	
SIGNATURE DATE	18

Output:

Addition: 15

Subtraction: 5

Multiplication: 50

Circle Area (radius = 7): 153.9380

Rectangle Area (5x10): 50

Triangle Area (base = 6, height = 8): 24.0